Time Trails

Gabriella Giannachi, University of Exeter, UK

Will Barrett, University of Exeter, UK

Rick Lawrence, Royal Albert Memorial Museum and Art Gallery, UK

Thomas Cadbury, Royal Albert Memorial Museum and Art Gallery, UK

Helen Burbage, Royal Albert Memorial Museum and Art Gallery, UK

Andy Chapman, 1010 Media, UK

Paul Farley, Exeter City Fc Supporters Trust, UK

Abstract: Time Trails is a collaboration between Royal Albert Memorial Museum and Art Gallery (RAMM), 1010 Media, University of Exeter, and the Exeter City FC Supporters' Trust (2013). The project researched the use of trails as a means to encounter and respond to RAMM's collections outside the museum in conjunction with content and branding provided by other partners, most prominently Exeter City FC Supporters' Trust. Drawing on research into trails (Ingold 2000 and 2007) and into trajectories as a way of designing mixed reality environments (Benford and Giannachi 2011), and focussing in particular on the orchestration of time, space, and roles, Time Trails investigated the curation of encounters with hybrid collections and archives via the creation of a number of chronological trails (e.g. Roman, Tudor, World War 2) and thematic trails (e.g. health and sport) that lead visitors through the city of Exeter. These trails can be further annotated by users via social media. Time Trails explored how users benefit from engaging creatively with hybrid collections outside the museum; how trails can be used by schools, the tourist industry, and local communities to raise cultural awareness and bring together diverse user groups. The framework developed from this project identifies the benefits of creative experiences of museum collections outside the museum, allowing organisations to engage audiences with objects which are not on display; learn how to stimulate creativity, reflection and self-documentation and facilitate free-style mobile learning in mixed reality environments; increase audience reach with social media to encourage participation among users who do not habitually visit museums; reconnect objects with the locations they were originally associated with and relate locations to different values; promote well-being by encouraging walking and collaborating with organisations that work with cultural tourists and local communities, including children who are at risk of disengaging from education.

Keywords: trails; digital heritage; mapping; trajectories; social media; mobile learning

Time Trails saw the Royal Albert Memorial Museum and Art Gallery (RAMM), 1010 Media, The Centre for Intermedia at the University of Exeter and Exeter City FC Supporters' Trust, collaborate to develop a web app prototype. This is based on RAMM's existing tours website, modelled on the Exeter Time Trail website, which is widely used by the BBC and Usborne, through which users can explore RAMM's collections via three kinds of tours: Museum tours, located at RAMM, linking various objects on display; Exeter tours, based on specific historical periods or themes; and Devon tours, linked to exhibitions and encouraging the exploration of Devon's wider heritage. Time Trails, which was developed in 2013, offers users a mobile learning experience allowing them to engage creatively with history, heritage, collections and archival materials through a series of time-based trails. These can be experienced both inside and outside the institutions that hosts them, whilst 'on the go', and include trails curated by RAMM (e.g., historic trails, such as our Roman, Tudor, Victorian, and a World War Two trails or thematic trails, such as life-style and sport trails), as well as trails curated by project partners, such as two Exeter City FC Supporters' Trust trails designed by Barrett and Giannachi in collaboration with staff from the Exeter City FC Supporters' Trust. Users can annotate these trails through social media and generate their own trails.

The Time Trails project drew on research into the use of trajectories for the design of mixed reality experiences (Benford and Giannachi 2011) and research into trails (Ingold 2000 and 2007) to investigate the value of following and generating trails to explore collection and archival materials outside of the institutions that host them. By mixed reality experiences we mean ones that span physical and digital environments (Milgram and Kishino 1994) and often entail forms of interactive,

distributed, and subjective performance (Benford and Giannachi 2011), in the sense that users of such environments are encouraged to perform actions or embrace forms of more or less minimal role-play. In particular, we utilised aspects of the trajectories framework, which was developed by Steve Benford and Gabriella Giannachi to provide a means to design and interpret mixed reality experiences (Benford and Giannachi 2011). For Benford and Giannachi trajectories define "predicted and actual itineraries through mixed reality experiences" (Benford and Giannachi 2011, p.15). The predicted, or canonic, trajectories express a desired set of routes participants may follow, and the actual, or participant, trajectories, show how participants choose to behave whilst on the go. We thought that the trajectories framework would allow us to plan how users may behave as they embrace a variety of more or less active roles (e.g. spectating, interacting, acting, documenting, doing, etc.) to deal with the shifts necessary in reflecting on and/or documenting one's experience of the overlaying of physical spaces with information about heritage received whilst on the go.

We decided to use maps for the visualisation of canonic trajectories and to facilitate orientation; mapping as a practice for knowledge generation that would be captured through self-documentation; and trails to encourage subjective forms of mobile learning. We used maps for orientation as we know they constitute 'graphic representations that facilitate a spatial understanding of things, concepts, conditions, processes or events in the human world' (Harley and Woodward 1987, p.xvi). We know that whilst maps are necessary orientation tools, they tend to express a canonic and holistic representation. We wanted to encourage users not so much to be passive receivers of knowledge, but rather to act as knowledge producers. We therefore wanted them not only to use maps for orientation but actively embrace the practice of mapping, which involves the production of knowledge 'on the go'. Reflections about this process could be captured through the use of social media which would also make possible the sharing of new knowledge to a broader user group. Finally, we used trails to facilitate subjective forms of mobile learning.

Our thinking in terms of the differences between these terms was influenced by the work of anthropologist Tim Ingold, who explains the distinctions between wayfinding, map-making and mapping that informed our understanding of the relationship between maps, mapping and trails. For Ingold wayfinding is 'the movement of people as they come and go between places' (2000, p.234) and mapping, the production of knowledge as one goes (2000, p.231), is a form of 're-enactment of those movements in inscriptive gestures' (2000, p.232). The map, he notes, captures neither, as it remains a representation. For him it is in fact wrong to assume that the 'structure of the map springs directly from the structure of the world' (2000, p.234). This ultimate 'carthographic illusion' (2000, p.234) has to do with the differences between networks and surfaces. In discussing a number of studies that show that Inuit people perceive their territory as a network of lines and itineraries rather than a surface (Collignon 1996, p.98; Aporta 2004, p.12), Ingold notes that while the British Royal Navy would follow specific points on route to a destination, moving 'across' a surface, the Inuit people moved 'along paths' (Ingold 2007, p.75). Whereas the map, which constitutes a means to facilitate transportation, can capture the former, the network of lines and paths formed by the latter remain too complex and subjective to be visualised on a map. We wanted to use maps for the purpose of navigation, and record predicted or canonic trajectories on them, but we also wanted to capture information about participant trajectories, documenting the subjective practice of mapping. This distinction between maps, mapping and trails formed the core of our framework. While maps or cartographies show a 'bird-eye's view' (Gibson 1979, pp.198-9), i.e. they show what claims to be an 'objective' reality, a representation that may be useful for transportation and orientation, trails show paths, making visible aspects of wayfinding, and allow the capture of mapping processes that document the unfolding of 'subjective' points of view. While maps represent more or less canonic and holistic world-views, useful for the telling of a history, trails constitute individual stories that document a subjective experience or path through a broader history, which may be represented by the map. Our vision for Time Trails was to build a tool that would allow the use of maps to capture what occurs whilst mapping, therefore empowering users to narrate individual stories in relation to and as part of a broader history.

Two projects played a significant role in shaping our thinking in terms of how walking could be used to stimulate mobile learning. Both adopted physical meandering as a performative strategy to change users' perception of places. These are Serendiptor and Dérive app. The former, developed by Mark Shepard at V2, combines a rooting service offered through Google Maps with instructions for action and movement inspired by Fluxus, Vito Acconci and Yoko Ono. For the Serendiptor team, the app will help users to 'find something by looking for something else'. Likewise, Dérive app, developed by a team led by Eduardo Cachucho, uses Situationist strategies to allow users to explore urban spaces 'in a care-free and casual way'. Interestingly, the Dérive app team postulate that 'active engagement of communities in their urban spaces unleashes in them new understanding of their urban surroundings',

hereby allowing users 'to see their urban spaces in a different light'. In line with these projects, we wanted to adopt physical meandering, which we knew through our research into Blast Theory's performance-work *Rider Spoke* (2007) can stimulate mental meandering (Benford and Giannachi 2011, p.188), to facilitate mapping and, through the production of knowledge involved in mapping, make users aware of their presence in relation to their changing relationship to places and the digital heritage that is associated with them. This led to the adoption, in one of our trails, the Exeter City FC Supporters' Trust trail, of basic tasks as a means to generate affordances that would encourage users to narrate their own memories in relation to their experience of a number of locations.

Other projects helped to shape our understanding of how to encourage users to attach their own memories to places. One such project is Findery, developed by a team in the San Francisco Bay Area, which allows users to leave notes, photos or videos to comment on specific destinations. Another such project is History Pin, developed by We Are What We Do in partnership with Google, which allows users to share stories about places. The History Pin has been adopted by a number of organisations which have used it in projects such as Olympic Memories, encouraging the gathering of photos and memories from the Olympics throughout the ages; Year of the Bay, capturing events occurring in the San Francisco Bay Area; and Putting Art on the Map, which facilitates the exploration of Imperial War Museum's collection of artworks on a Google Map. All these projects show that pinpointing memories and stories to locations, and using maps as a means to prompt the telling of personalized stories in relation to these locations, is an increasingly popular strategy to produce engagement with local history. Most crucially, they show the value of mapping, generating trails and even map-making as strategies for remembering.

Another project utilizing mapping as well as map-making to reflect about place and art is Artmaps, a collaboration between the Centre for Intermedia at the University of Exeter, University of Nottingham and three departments at Tate, Tate Online, Tate Research and Tate Learning, which aimed to give users the possibility of exploring artworks in the Tate collection in relation to the places, sites, landscapes and environments that informed or led to their geotagging. The web app, which also uses a map as interface, can locate their user and bring up works in the Tate collection that are geotagged in relation to places near them. Users could then look at these works on the map and/or explore them in situ, reflecting on how what they see in the works relates to their surroundings. Alternatively, through a search function, they could explore works in any locality. The original aim of the project was for the apps to facilitate crowdsourcing so that Tate's knowledge of the artworks' geolocation could be improved on. Additional functionalities now prompt users to self-document their encounter with the work through text, photography, video and sound (see Cardiff, Sinker and Beaven 2013), allowing Tate to build a growing understanding of how users interpret artworks in their collection whilst on the go. Also influential on the development of Time Trails as a tool to reflect about artifacts and art *in situ* was Moor Stories, a collaboration between the Centre for Intermedia at the University of Exeter, RAMM and 1010 Media, which generated a web app that aims to facilitate creative encounters between objects in RAMM's collections pertaining to Dartmoor (primarily flints, pottery, drawings, paintings, wood and stone carvings) to prompt the documentation of memories about them among the communities that live where the objects were originally found. Whilst Moor Stories was more specifically oriented towards the telling of stories (whether creative, fictional or factual), it still offered a valuable model to understand how to prompt users to reflect about a particular location in relation to an artifact or artwork. Both Art Maps and Moor Stories projects allow users to contribute individual knowledge to a canonic interpretation and, by using social media, to share this knowledge with broader user communities. Both also make it possible for Tate and RAMM to present users with novel ways of encountering their collections, including items not ordinarily on display, and facilitate more creative and performative engagements. Finally both projects 'exploded' the idea of the museum as a public space by physically moving the exhibition space, alongside its interpretation, outside of the museum and distributing within users' everyday life environments.

Finally we looked into the operation of a number of platforms which facilitate trail creation, such as viewranger, allowing users to generate and share trails primarily to talk about journeys and experiences, and EveryTrail, allowing users to generate trails, and add texts and images to them. Other apps that specifically use trails to facilitate the exploration of places through art and history include 7 scenes whose project website notes that as 'the smartphone is quickly becoming our primary access to cultural information', people increasingly 'use apps on their phone to find cultural landmarks, to take tours exploring the city, to learn more during exhibitions and to share experiences' in order to 'experience cultural heritage' in novel ways. 'Tourmaker' is another such tool which allows users to select a genre which determines if the user can navigate freely or is restricted in any way; produce a story, which is meant to generate expectation; choose a configuration, which determines a tour's time

limit; specify localities, dragging and dropping texts, photos and sounds; and finally publish and share their tour. 1001 Stories about Denmark, likewise, aims to collect stories from the Iron Age settlement Sorte Muld ('Black Soil') on Bornholm to the Black Horse Cavern in Copenhagen in the form of trails which can be walked, cycled or driven by car. In this case the histories behind the trails are written by historians and heritage experts, though they can then also be annotated by users. What interested us about these projects was the idea that a Time Trails app could act as an orientation tool, a search tool, and a publishing tool to tie canonic trajectories (or locative stories created by experts) to participant trajectories (documentations of engagement with these stories produced by users).

Following a period of research and development with our stakeholders, which saw us consult with two primary schools in Exeter (St Sidwell's Primary School and St David's Primary School), Exeter City Red Coats Guides, and various members of the Exeter City FC Supporters' Trust, 1010 Media developed the Time Trails web app together with a "find me" tool that allows users to identify their location by utilising geocoding so that their location may be imported into a trail and become a destination. Once a tour is selected, the tool identifies the user's location through GPS and places them on the map as a green dot, if connectivity is good, or an orange dot if it isn't. If the user is in the proximity of a trail, they can start it and progress through as many locations as they have time for. These are numbered to give the idea of a trail, which implies direction and sequencing. Alternatively, the app suggests that the user views the locations as a list. Once a location is encountered, the user can view it, move on or comment about it on twitter. Once a location has been viewed the icon on the map that represents the location turns from red to green, though these data can be erased so as to take the tour again. The interface, which initially showed a map view, was changed to a hybrid view to create an enhanced sense of being in a location. Street view can also be used, and all settings can be erased so as to experience the tour afresh.

A number of trails used content from RAMM's collections (such as a Tudor trail to coincide with an exhibition on Elizabethan art scheduled for October 2013 and a World War Two trail) and two utilized Exeter City FC Supporters' Trust content (one was written for the general public and one was written specifically for the children taking part in the Football in the Community Kick Start programme which supports children who are at risk of disengaging from education by delivering a personalized learning programme). The RAMM trails used the existing trails of the Exeter Time Trail website, with the exception of the World War Two trail which was designed by Will Barrett specifically for this project. Both Exeter City FC Supporters' Trust trails were designed by asking stakeholders from Exeter City FC Supporters' Trust to identify a set of locations in Exeter and at St James Park, the stadium that is home to Exeter City Football Club, through which one could narrate the history of both the Club and the Trust that owns it. Whereas in the RAMM trails locations are tied to where objects in RAMM's collections were initially found, the locations forming part of the Exeter City FC Supporters' Trust trails are tied to oral histories, only occasionally, referring to artefacts by which to remember them. Each of the interpretation texts of the longer Exeter City FC Supporters' Trust trail is meant to prompt an affordance facilitating users' in relating themselves to the physical sites and the stories narrated through them, and thus position themselves within the broader history of the Club and Trust in Exeter whereas the texts in the shorter trail, written for the children taking part in the Kick Start programme, did not contain affordances and feedback was gathered at the end of a structured mobile learning programme through a role play exercise that saw the children trying to persuade a member of the team to sign up for Exeter City FC.

Below are examples of texts from the Tudor trail, the World War Two trail, and the Exeter City FC Supporters' Trust longer trail that refer to a set of locations in proximity to each other:

> **Doors from the High School:** These complex doors of oak came from the Exeter Free Grammar School, which stood on the High Street close to the Eastgate. The site is now occupied by River Island. With their radiating 'shell' motif head, row of face-masks and stud-like motifs, the doors are especially similar to the door of Nos 10-11 Cathedral Close, and more generally to Nicholas Baggett's door to the Guildhall. It has been claimed that they date to 1593 but they may equally belong to the foundation of the Grammar School in 1633.
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> **Exeter Cathedral and the Close** by William Lionel Clause (1943) (© Royal Albert Memorial Museum & Art Gallery, Exeter): Presented by HM Government War Artists' Advisory Committee. The Cathedral escaped the bombing with only limited damage in 1942, but many of the buildings of the Cathedral Close and South Street were demolished or badly damaged. This pen-and-watercolour view by the war artist W.L. Clause looks towards the Close, showing the state of the Cathedral and Close following extensive demolitions of badly damaged buildings.

> **Jimmy Rigby shop:** Jimmy Rigby (b. 1886) was part of the side that travelled to South America in 1914. Exeter City was selected to do so by the Southern League that felt they represented a 'typical English football team', the tour contained a number of memorable games and incidents; none more so than the 2-0 victory over Argentine Champions, Racing Club de Buenos Aires, a match which had to be temporarily abandoned when the secretary of the South American side brandished his revolver towards the ref in frustration at City taking the lead! Jimmy Rigby, who early in his career combined playing whilst working as a cotton spinner in the textile industry, later owned three shops in this area which were run by various members of his family. There was one here at Sidwell Street, near where Sainsburys currently is, one at the top of Paris Street, and one near the Old Tiverton Road roundabout. The shop here and the one near the Old Tiverton Road roundabout were destroyed during the 1942 Blitz on Exeter. Rigby was a director of Exeter City for many years.

It is clear from these examples that the texts aim to offer interpretation about artefacts, in case of RAMM's trails, or offer a perspective over a broader history, in case of the Exeter City FC Supporters' Trust trail, whilst drawing attention to the discrepancies between the past and present uses of these sites and, possibly, interpretations of historically significant moments according to different points of view. Examples such as the one below, tied to the football stadium where Exeter City FC play, drawn from the Exeter City FC Supporters' Trust trail, also reveal how the team attempted to prompt user engagement through tasks, affordances and questions:

> Exeter City FC's first match was held here at St James Park on 10th September 1904. The team won 2-1 against the 110th First Royal Artillery. The winning goal was scored by Sid Thomas, whose career with the club would go on to last over 70 years not only as a player, but also a secretary, director, chairman and lifelong president. When ECFC turned professional in 1908, St James Park was developed to meet the standards of the Southern League. Despite a few games played elsewhere, this has been the home of Exeter City FC for over 100 years. Once there was a large gate here, the Kendall Gate. On a Friday in late February 1981, when we had an FA cup run, hundreds of supporters slept overnight on the pavement by that gate, so when the shop opened on the Saturday morning to sell tickets for the quarter final match away to Spurs we would not miss out. Were you there that night? Can you remember what you felt the first time you came here?

Whilst these locations form part of different trails, each written according to a specific chronology or history, brought together, they generate a hybrid 'live' map visualising the changing history of Exeter over time in the eyes of diverse communities.

We hope to have shown how the combined use of maps for the visualisation of canonic trajectories and for orientation, mapping as a practice for knowledge generation that would be captured through reflection and self-documentation, and shared via social media, and trails to encourage subjective forms of mobile learning, has created a novel way to not only encounter and interpret but also write individual histories into the broader history of a community. Initial testing of both RAMM and Exeter City FC Supporters' Trust trails showed that, as in the case of the Art Maps project, users who encountered collections outside of the museum frequently wished to go back to the museum to encounter the 'original' artefacts. This suggests that Time Trail offers a novel way to 'exhibit' artefacts, in the case of RAMM, and locate and write up lost or misplaced histories, in the case of Exeter City FC Supporters' Trust. Using the content of the Exeter City FC Supporters' Trust trail in the context of the Kick Start programme also proved a valid way to stimulate creativity and reflection whilst facilitating free-style mobile learning among children who may have become disaffected with conventional learning. This, alongside the use of social media to disseminate knowledge produced on the go, allows organisations to increase audience reach and encourage reflection about digital heritage among users who do not habitually visit museums or may have disengaged from learning. In all cases, Time Trails makes it possible for users to reconnect objects with the locations they were 'originally' associated with, thus allowing them to re-engage with objects and histories that had been taken away from them, changing their perception of locations encountered in everyday life through the embedding of information that could increase digital heritage's aesthetic and emotional value for diverse kinds of user communities.

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References

Aporta, C. (2004) 'Routes, trails and tracks: trail breaking among the Inuit of Igloolik', Études/Inuit/Studies, 28(2): 9-38.

Art Maps, http://www.tate.org.uk/about/projects/art-maps, Accessed 6 October 2013.

Benford, S. and Giannachi, G. (2011) *Performing Mixed Reality*, Cambridge, Mass.: The MIT Press.

Cardiff, R., Sinker, R., and Beaven, K. (2013) 'Taking the Collection out of the Gallery', *Museums and the Web,* April 17-20, Portland, Oregon, Available at http://mw2013.museumsandtheweb.com/paper/taking-the-collection-out-of-the-gallery/, Accessed 6 October 2013.

Collignon, B. (1996) Les Inuit: Ce qu'ils savent du territoire, Paris: L'Hartmattan.

Dérive app, http://deriveapp.com/s/v2/, Accessed 7 October 2013.

EveryTrail, http://www.everytrail.com/, Accessed 8 October 2013.

Findery, http://findery.com, Accessed 7 October 2013.

Grecian Voices, http://grecianvoices.wordpress.com/, Accessed 26 September 2013.

Gibson, J. (1979) *The Ecological Approach to Visual Perception***,** Boston: Houghton Mifflin.

Harley, J. B. and Woodward, D. (1987) Eds. *The History of Geohraphy, vol. 1,* Chicago. Chicago University Press.

History Pin, http://www.historypin.com/, Accessed 7 October 2013.

Ingold, T. (2000) *The Perception of the Environment,* London and New York: Routledge.

Ingold, T. (2007) *Lines: a brief history*, London and New York: Routledge.

Milgram, P. and Kishino, F. (1994) 'A taxonomy of mixed reality displays'. *IEICE Transactions on Information Systems.* E77-D12, 449-455.

Moor Stories, http://moorstories.org.uk/, Accessed 7 October 2013.

Serendiptor, http://serendipitor.net/site/, Accessed 7 October 2013.

7 Scenes, http://7scenes.com/, Accessed 7 October 2013.

1001 Stories about Denmark,

http://www.kulturarv.dk/1001fortaellinger/en_GB/routes, Accessed 7 October 2013.

Viewranger, http://my.viewranger.com/route/details/MTk5MzY, Accessed 7 October 2013.